REBAL | ViT feature adapters

Matched frozen-encoder study on Tiny ImageNet
Experimental report - 12 October 2026

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Completed and independently verified: 30 paired condition-by-seed cells, 90 trained feature adapters, 270 retrained linear classifiers and 810 named readouts. Both pretrained image encoders remain frozen. Every adapter completed 50 epochs and every retrained classifier completed 2,000 updates.

Primary encoder effects

Paired contrast	Delta (pp)	95% t interval	t p, Holm	Exact p, Holm
10:1 | DeiT - ResNet	+8.85	[+8.24, +9.46]	3.13e-05	0.938
50:1 | DeiT - ResNet	+12.58	[+11.49, +13.67]	7.30e-05	0.938
100:1 | DeiT - ResNet	+13.42	[+12.53, +14.30]	2.85e-05	0.938

The encoder contrast compares DeiT-tiny with ResNet18 using CB+EQ adapters and streaming-SMOTE classifiers. Ratio labels denote nominal rho; integer rounding changes the realized maximum/minimum class-count ratio. Deltas are paired macro-F1 percentage points across five seeds, with unadjusted 95% t intervals. Negative values favor ResNet18. All 15 prespecified comparisons share one Holm family for each test type.

Absolute control performance

Encoder / ratio	Native CE macro-F1	CE + LA macro-F1	CB+EQ + SMOTE macro-F1
ResNet18 | 10:1	52.79 +/- 0.59	56.38 +/- 0.22	55.55 +/- 0.41
ResNet18 | 50:1	38.44 +/- 0.82	49.27 +/- 0.58	46.54 +/- 0.63
ResNet18 | 100:1	32.15 +/- 0.46	45.20 +/- 0.52	41.48 +/- 0.56
DeiT-tiny | 10:1	62.07 +/- 0.34	64.93 +/- 0.32	64.40 +/- 0.14
DeiT-tiny | 50:1	49.44 +/- 0.69	60.11 +/- 0.48	59.12 +/- 0.63
DeiT-tiny | 100:1	41.47 +/- 0.51	56.37 +/- 0.37	54.90 +/- 0.91

DeiT has large positive paired encoder effects under the requested configuration, but that pipeline is not dominant: CE + LA has higher mean macro-F1 in all six conditions (by 0.526-3.722 points). These absolute-control comparisons are descriptive and outside the 15-member primary family. All 27 readouts remain in the complete tables.

CE + LA does not dominate all metrics: for DeiT at nominal 100:1, CB+EQ + SMOTE tail-class F1 is 45.98% versus 42.93% for CE + LA. This descriptive tail-versus-overall tradeoff is outside the primary inference family.

The six EQ-minus-CB condition means are all negative and small (-0.0201 to -0.0007 percentage points), without establishing harm or equivalence. SMOTE-minus-ROS signs differ by encoder and ratio, with no consistent benefit and no Holm-corrected primary evidence of improvement. Negative results and seed uncertainty remain visible.

Scope: frozen ImageNet-pretrained encoders plus trained adapters, not end-to-end ViT learning. This study tests no GAN and provides no causal estimate of replacing a GAN. ImageNet/Tiny ImageNet overlap limits unseen-image generalization claims. Five paired seeds cannot yield a two-sided exact sign-flip p below 0.0625.

1. Data, scale and matched preprocessing

Source / split	Images	Meaning
Official training	100,000	200 classes; 500 source images per class
Clean development	99,973	Training/evaluation pixel overlap and repeated training pixels excluded
Calibration per seed	10,000	50 retained training images per class
Final evaluation	10,000	Official validation images; 50 per class
Fitted, nominal 10:1	34,765	445 down to 44 examples per class
Fitted, nominal 50:1	22,309	445 down to 8 examples per class
Fitted, nominal 100:1	19,168	445 down to 4 examples per class

The source rows and labels are identical for both encoders. Before splitting, training pixels matching evaluation pixels are removed; the first remaining training image for each exact pixel hash is retained. This local duplicate policy removes 27 training rows and leaves evaluation unchanged. It does not detect semantic duplicates or pretraining exposure.

Seeds 0-4 control independent calibration, semantic-rank and fit-selection streams. Each seed has nested nominal rho 10/50/100 fit sets with counts floor(445 x rho^(-rank/199)). Flooring gives realized maximum/minimum ratios 445/44 = 10.11:1, 445/8 = 55.625:1, and 445/4 = 111.25:1. All table and plot ratio labels denote nominal rho. Frequency regions contain 66 head, 67 middle and 67 tail classes. Within a seed and ratio, both encoders and all losses use exactly the same split, row order and semantic rank.

One deterministic 224-pixel input convention

Original 64 x 64 RGB images are resized to 256 using PIL bicubic interpolation, center-cropped to 224, scaled to [0,1], and normalized with ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). Feature extraction uses no random augmentation. Enlarging images does not create higher-resolution observations.

This shared preprocessing follows the DeiT evaluation convention [1,2]. ResNet18 V1 normally uses bilinear resizing [3]; bicubic is an intentional matched-input control, not a claim of each model's individually optimal preprocessing. DeiT patch size 16 gives 196 image patches at 224, without adapting its pretrained positional grid. The earlier native64 ResNet transfer study is a separate experiment.

2. Frozen encoders and trainable controls

Component	ResNet18	DeiT-tiny/16
Embedding dimensions	512	192
Frozen backbone parameters	11,176,512	5,524,416
Adapter parameters	164,224	82,304
Each native/retrained head	25,800	25,800

ResNet18 uses torchvision IMAGENET1K_V1 weights. DeiT uses the official non-distilled tiny model, pretrained on ImageNet-1K [1-3]. Original 1,000-class heads are removed. Neither frozen backbone receives downstream gradient updates. Different architecture, capacity, feature dimension and pretraining recipe remain coupled; this compares encoder packages, not architecture in isolation.

Component	Fixed downstream design
Input scaling	Per-dimension mean/std fitted only on retained fit features, separately for each encoder/cell.
Adapter	InputD -> 256 -> 128; ReLU/dropout 0.1; native 200-class linear head.
Objectives	CE; deferred class-balanced CE (CB); deferred CB plus equalization (CB+EQ).
Optimization	50 epochs, batch 1,024, LR 0.05, five warmup epochs then cosine decay; Nesterov SGD momentum 0.9.
Regularization	Linear L2 0.0005; per-tensor gradient clipping 1; EQ coefficient 0.05; outlier filtering off.
Deferred controls	CB/EQ active in final 20% of epochs; beta=1-1/n_max; expected fit-example CB weight 1.

All three losses share complete adapter initialization and batch order within an encoder cell. Across encoders, input projection dimensions differ, so complete adapter initialization cannot be identical. Native and retrained 128-to-200 heads share the same initialization per seed across encoders. EQ modifies the learned 128D adapter representation, never the pretrained encoder.

The repository's EQ loss normalizes features, reduces within-class squared radius and subtracts mean pairwise squared distance of eligible class centers. A class needs at least two examples in a batch; separation needs at least two eligible classes. The CB beta/normalization and deferred schedule adapt effective-number weighting [4]; this is not a complete two-factor loss ablation.

3. Classifier rebalancing and readout definitions

Readout	Definition
Native	Classifier learned jointly with its CE, CB or CB+EQ adapter.
Natural retraining	Freeze adapter; sample fitted examples uniformly.
ROS retraining	Freeze adapter; sample classes uniformly, then real examples within class.
Streaming latent SMOTE	Same class-uniform prior as ROS, with fresh same-class interpolation.
NCM	Nearest normalized fitted class centroid; exploratory geometry readout.
Logit adjustment	Calibration-selected tau on a fixed grid; objective-specific training prior.

Following decoupled representation/classifier training [5], each of the 90 adapters receives three separately fitted linear heads: natural, ROS and streaming latent SMOTE. All 270 heads use 2,000 updates of batch 2,048, SGD LR 0.01, momentum 0.9 and linear L2 0.0005. Equal update counts do not imply equal natural-example exposure across imbalance ratios.

SMOTE adapts within-class interpolation [6] to fitted latent vectors. Choose a class uniformly, a real anchor uniformly, and an exact same-class neighbor excluding the anchor; interpolate with a fresh uniform coefficient with probability 1-n_class/n_max. Maximal classes use real samples only. This streaming variant differs from a finite static synthetic pool and does not generate pixel images.

Post-hoc logit adjustment [7] selects tau using calibration macro-F1, with ties resolved by the declared smallest-tau convention. The native CB/CB+EQ prior approximates the final weighted objective despite deferred training. Natural heads use empirical fit priors. ROS/SMOTE priors are uniform, making their tau-zero LA views identical to base predictions.

Count	Interpretation
30 cells	2 encoders x 3 ratios x 5 seeds
90 trained adapters	3 objective fits per cell
270 retrained classifiers	3 samplers per adapter
810 named readouts	27 views per cell; includes 180 uniform-prior LA duplicates
450 stored base readouts	15 per cell; remaining LA views reconstruct from saved prior/tau
360 calibration curves	12 per cell; no test-selected adjustment

The final epoch and fixed head update budget are evaluated; test scores select no checkpoint, architecture or schedule. The vectorized sampler removes a Python class loop while preserving all five RNG streams, realized samples and classifier trajectories in regression checks.

4. Complete results: ResNet18 | 10:1

Mean +/- sample SD across five paired seeds; values are percentages. Accuracy, macro-F1, macro recall and head/middle/tail macro-F1 are reported for every declared readout. Macro recall equals accuracy on this class-balanced evaluation set.

Method / readout	Accuracy	Macro F1	Macro recall	Head F1	Middle F1	Tail F1
CE	53.52 +/- 0.58	52.79 +/- 0.59	53.52 +/- 0.58	55.20 +/- 1.71	54.88 +/- 0.84	48.31 +/- 1.43
CE + LA	56.57 +/- 0.22	56.38 +/- 0.22	56.57 +/- 0.22	60.69 +/- 2.00	56.28 +/- 1.01	52.25 +/- 1.53
CE / NCM	53.69 +/- 0.14	53.26 +/- 0.17	53.69 +/- 0.14	56.73 +/- 1.93	52.79 +/- 0.77	50.30 +/- 1.42
CE / natural	53.14 +/- 0.50	52.51 +/- 0.44	53.14 +/- 0.50	54.97 +/- 1.86	54.62 +/- 0.93	47.96 +/- 1.05
CE / natural + LA	55.92 +/- 0.31	55.86 +/- 0.31	55.92 +/- 0.31	60.54 +/- 1.70	55.86 +/- 1.40	51.24 +/- 1.42
CE / ROS	55.89 +/- 0.26	55.64 +/- 0.30	55.89 +/- 0.26	59.12 +/- 1.98	55.60 +/- 1.09	52.26 +/- 1.75
CE / ROS + LA	55.89 +/- 0.26	55.64 +/- 0.30	55.89 +/- 0.26	59.12 +/- 1.98	55.60 +/- 1.09	52.26 +/- 1.75
CE / SMOTE	55.74 +/- 0.42	55.53 +/- 0.46	55.74 +/- 0.42	58.83 +/- 2.09	55.57 +/- 1.01	52.23 +/- 1.76
CE / SMOTE + LA	55.74 +/- 0.42	55.53 +/- 0.46	55.74 +/- 0.42	58.83 +/- 2.09	55.57 +/- 1.01	52.23 +/- 1.76
CB	56.41 +/- 0.26	56.02 +/- 0.30	56.41 +/- 0.26	58.71 +/- 1.85	56.13 +/- 0.75	53.26 +/- 1.66
CB + LA	56.75 +/- 0.25	56.49 +/- 0.26	56.75 +/- 0.25	60.07 +/- 1.98	56.25 +/- 1.02	53.21 +/- 1.61
CB / NCM	53.78 +/- 0.25	53.37 +/- 0.27	53.78 +/- 0.25	56.77 +/- 2.00	52.83 +/- 0.70	50.56 +/- 1.43
CB / natural	53.24 +/- 0.46	52.64 +/- 0.38	53.24 +/- 0.46	54.98 +/- 1.84	54.71 +/- 0.95	48.27 +/- 1.12
CB / natural + LA	56.02 +/- 0.31	55.99 +/- 0.31	56.02 +/- 0.31	60.64 +/- 1.88	55.92 +/- 1.30	51.48 +/- 1.43
CB / ROS	55.92 +/- 0.26	55.68 +/- 0.32	55.92 +/- 0.26	59.15 +/- 1.97	55.60 +/- 1.07	52.34 +/- 1.81
CB / ROS + LA	55.92 +/- 0.26	55.68 +/- 0.32	55.92 +/- 0.26	59.15 +/- 1.97	55.60 +/- 1.07	52.34 +/- 1.81
CB / SMOTE	55.78 +/- 0.37	55.57 +/- 0.38	55.78 +/- 0.37	58.71 +/- 2.10	55.63 +/- 0.92	52.42 +/- 1.61
CB / SMOTE + LA	55.78 +/- 0.37	55.57 +/- 0.38	55.78 +/- 0.37	58.71 +/- 2.10	55.63 +/- 0.92	52.42 +/- 1.61
CB+EQ	56.43 +/- 0.25	56.03 +/- 0.28	56.43 +/- 0.25	58.70 +/- 1.88	56.15 +/- 0.70	53.29 +/- 1.61
CB+EQ + LA	56.73 +/- 0.26	56.47 +/- 0.26	56.73 +/- 0.26	60.05 +/- 1.99	56.19 +/- 1.02	53.21 +/- 1.59
CB+EQ / NCM	53.73 +/- 0.27	53.31 +/- 0.28	53.73 +/- 0.27	56.70 +/- 2.01	52.78 +/- 0.77	50.50 +/- 1.38
CB+EQ / natural	53.23 +/- 0.46	52.63 +/- 0.38	53.23 +/- 0.46	54.94 +/- 1.82	54.72 +/- 0.95	48.25 +/- 1.14
CB+EQ / natural + LA	56.02 +/- 0.36	55.99 +/- 0.28	56.02 +/- 0.36	60.66 +/- 1.79	55.91 +/- 1.25	51.46 +/- 1.36
CB+EQ / ROS	55.91 +/- 0.24	55.67 +/- 0.31	55.91 +/- 0.24	59.14 +/- 1.98	55.60 +/- 1.06	52.33 +/- 1.76
CB+EQ / ROS + LA	55.91 +/- 0.24	55.67 +/- 0.31	55.91 +/- 0.24	59.14 +/- 1.98	55.60 +/- 1.06	52.33 +/- 1.76
CB+EQ / SMOTE	55.76 +/- 0.40	55.55 +/- 0.41	55.76 +/- 0.40	58.72 +/- 2.10	55.59 +/- 0.95	52.38 +/- 1.63
CB+EQ / SMOTE + LA	55.76 +/- 0.40	55.55 +/- 0.41	55.76 +/- 0.40	58.72 +/- 2.10	55.59 +/- 0.95	52.38 +/- 1.63

CB+EQ identifies the REBAL adapter. /natural, /ROS and /SMOTE are retrained linear heads on that adapter. NCM uses fitted centroids. LA is a view of the same classifier, not another fit; balanced-head duplicate rows are deliberate. Zero-denominator class metrics use zero.

The accompanying CSVs retain every seed-level scalar (including worst-class F1 and head-tail gap), all per-class precision/recall/F1 values, the six-metric summaries, and all primary comparisons. No favorable subset replaces this complete table.

5. Complete results: ResNet18 | 50:1

Mean +/- sample SD across five paired seeds; values are percentages. Accuracy, macro-F1, macro recall and head/middle/tail macro-F1 are reported for every declared readout. Macro recall equals accuracy on this class-balanced evaluation set.

Method / readout	Accuracy	Macro F1	Macro recall	Head F1	Middle F1	Tail F1
CE	42.22 +/- 0.43	38.44 +/- 0.82	42.22 +/- 0.43	47.90 +/- 1.02	46.65 +/- 1.12	20.91 +/- 2.48
CE + LA	48.66 +/- 0.85	49.27 +/- 0.58	48.66 +/- 0.85	59.10 +/- 1.84	50.61 +/- 1.45	38.26 +/- 2.00
CE / NCM	47.70 +/- 0.58	46.65 +/- 0.72	47.70 +/- 0.58	54.07 +/- 1.46	47.43 +/- 0.98	38.58 +/- 2.17
CE / natural	42.20 +/- 0.43	38.63 +/- 0.85	42.20 +/- 0.43	48.14 +/- 1.09	46.40 +/- 1.32	21.49 +/- 2.43
CE / natural + LA	48.00 +/- 0.51	48.85 +/- 0.47	48.00 +/- 0.51	58.75 +/- 1.99	50.33 +/- 1.41	37.63 +/- 1.85
CE / ROS	47.97 +/- 0.66	46.66 +/- 0.74	47.97 +/- 0.66	54.81 +/- 1.77	48.60 +/- 1.32	36.68 +/- 1.89
CE / ROS + LA	47.97 +/- 0.66	46.66 +/- 0.74	47.97 +/- 0.66	54.81 +/- 1.77	48.60 +/- 1.32	36.68 +/- 1.89
CE / SMOTE	47.86 +/- 0.57	46.55 +/- 0.63	47.86 +/- 0.57	54.21 +/- 1.75	48.76 +/- 1.33	36.78 +/- 1.61
CE / SMOTE + LA	47.86 +/- 0.57	46.55 +/- 0.63	47.86 +/- 0.57	54.21 +/- 1.75	48.76 +/- 1.33	36.78 +/- 1.61
CB	48.78 +/- 0.57	47.41 +/- 0.67	48.78 +/- 0.57	54.48 +/- 1.48	49.66 +/- 1.18	38.18 +/- 1.88
CB + LA	49.55 +/- 0.52	48.62 +/- 0.63	49.55 +/- 0.52	56.69 +/- 1.60	49.93 +/- 1.17	39.35 +/- 2.04
CB / NCM	47.88 +/- 0.48	46.89 +/- 0.64	47.88 +/- 0.48	54.09 +/- 1.44	47.44 +/- 1.10	39.26 +/- 2.06
CB / natural	42.33 +/- 0.57	38.91 +/- 0.95	42.33 +/- 0.57	48.06 +/- 1.04	46.44 +/- 1.48	22.38 +/- 2.51
CB / natural + LA	48.43 +/- 0.66	49.09 +/- 0.53	48.43 +/- 0.66	58.73 +/- 1.95	50.37 +/- 1.28	38.31 +/- 2.07
CB / ROS	47.95 +/- 0.66	46.60 +/- 0.73	47.95 +/- 0.66	54.56 +/- 1.85	48.65 +/- 1.30	36.72 +/- 1.80
CB / ROS + LA	47.95 +/- 0.66	46.60 +/- 0.73	47.95 +/- 0.66	54.56 +/- 1.85	48.65 +/- 1.30	36.72 +/- 1.80
CB / SMOTE	47.87 +/- 0.58	46.54 +/- 0.64	47.87 +/- 0.58	54.09 +/- 1.72	48.81 +/- 1.37	36.83 +/- 1.56
CB / SMOTE + LA	47.87 +/- 0.58	46.54 +/- 0.64	47.87 +/- 0.58	54.09 +/- 1.72	48.81 +/- 1.37	36.83 +/- 1.56
CB+EQ	48.79 +/- 0.56	47.42 +/- 0.66	48.79 +/- 0.56	54.49 +/- 1.50	49.67 +/- 1.17	38.20 +/- 1.86
CB+EQ + LA	49.54 +/- 0.51	48.60 +/- 0.63	49.54 +/- 0.51	56.67 +/- 1.60	49.89 +/- 1.13	39.36 +/- 2.04
CB+EQ / NCM	47.83 +/- 0.49	46.84 +/- 0.65	47.83 +/- 0.49	54.05 +/- 1.45	47.40 +/- 1.11	39.18 +/- 2.07
CB+EQ / natural	42.33 +/- 0.58	38.91 +/- 0.97	42.33 +/- 0.58	48.06 +/- 1.03	46.44 +/- 1.50	22.37 +/- 2.56
CB+EQ / natural + LA	48.41 +/- 0.67	49.07 +/- 0.53	48.41 +/- 0.67	58.73 +/- 1.94	50.37 +/- 1.28	38.27 +/- 2.09
CB+EQ / ROS	47.98 +/- 0.66	46.63 +/- 0.73	47.98 +/- 0.66	54.57 +/- 1.84	48.67 +/- 1.28	36.75 +/- 1.82
CB+EQ / ROS + LA	47.98 +/- 0.66	46.63 +/- 0.73	47.98 +/- 0.66	54.57 +/- 1.84	48.67 +/- 1.28	36.75 +/- 1.82
CB+EQ / SMOTE	47.87 +/- 0.58	46.54 +/- 0.63	47.87 +/- 0.58	54.12 +/- 1.69	48.78 +/- 1.42	36.83 +/- 1.52
CB+EQ / SMOTE + LA	47.87 +/- 0.58	46.54 +/- 0.63	47.87 +/- 0.58	54.12 +/- 1.69	48.78 +/- 1.42	36.83 +/- 1.52

CB+EQ identifies the REBAL adapter. /natural, /ROS and /SMOTE are retrained linear heads on that adapter. NCM uses fitted centroids. LA is a view of the same classifier, not another fit; balanced-head duplicate rows are deliberate. Zero-denominator class metrics use zero.

The accompanying CSVs retain every seed-level scalar (including worst-class F1 and head-tail gap), all per-class precision/recall/F1 values, the six-metric summaries, and all primary comparisons. No favorable subset replaces this complete table.

6. Complete results: ResNet18 | 100:1

Mean +/- sample SD across five paired seeds; values are percentages. Accuracy, macro-F1, macro recall and head/middle/tail macro-F1 are reported for every declared readout. Macro recall equals accuracy on this class-balanced evaluation set.

Method / readout	Accuracy	Macro F1	Macro recall	Head F1	Middle F1	Tail F1
CE	37.70 +/- 0.58	32.15 +/- 0.46	37.70 +/- 0.58	45.35 +/- 1.22	41.98 +/- 1.45	9.33 +/- 1.29
CE + LA	44.15 +/- 0.55	45.20 +/- 0.52	44.15 +/- 0.55	58.01 +/- 2.25	48.16 +/- 1.05	29.62 +/- 1.98
CE / NCM	44.24 +/- 0.59	42.44 +/- 0.58	44.24 +/- 0.59	52.53 +/- 1.52	44.52 +/- 1.02	30.43 +/- 1.42
CE / natural	37.83 +/- 0.58	32.54 +/- 0.43	37.83 +/- 0.58	45.55 +/- 1.26	41.92 +/- 1.36	10.36 +/- 1.11
CE / natural + LA	43.80 +/- 0.59	44.95 +/- 0.54	43.80 +/- 0.59	57.73 +/- 2.42	47.98 +/- 1.11	29.33 +/- 1.88
CE / ROS	44.06 +/- 0.47	41.60 +/- 0.59	44.06 +/- 0.47	52.39 +/- 1.52	45.45 +/- 1.30	27.11 +/- 1.72
CE / ROS + LA	44.06 +/- 0.47	41.60 +/- 0.59	44.06 +/- 0.47	52.39 +/- 1.52	45.45 +/- 1.30	27.11 +/- 1.72
CE / SMOTE	43.88 +/- 0.46	41.41 +/- 0.59	43.88 +/- 0.46	51.79 +/- 1.56	45.57 +/- 1.15	27.03 +/- 1.75
CE / SMOTE + LA	43.88 +/- 0.46	41.41 +/- 0.59	43.88 +/- 0.46	51.79 +/- 1.56	45.57 +/- 1.15	27.03 +/- 1.75
CB	44.77 +/- 0.44	42.35 +/- 0.56	44.77 +/- 0.44	52.36 +/- 1.41	46.55 +/- 1.28	28.28 +/- 1.60
CB + LA	45.74 +/- 0.44	43.85 +/- 0.56	45.74 +/- 0.44	54.64 +/- 1.60	46.97 +/- 1.38	30.10 +/- 1.78
CB / NCM	44.52 +/- 0.67	42.84 +/- 0.72	44.52 +/- 0.67	52.59 +/- 1.56	44.75 +/- 1.27	31.32 +/- 1.59
CB / natural	38.02 +/- 0.56	32.88 +/- 0.49	38.02 +/- 0.56	45.68 +/- 1.13	42.04 +/- 1.29	11.11 +/- 1.01
CB / natural + LA	44.19 +/- 0.60	45.32 +/- 0.56	44.19 +/- 0.60	57.64 +/- 2.24	48.01 +/- 1.24	30.49 +/- 1.92
CB / ROS	44.09 +/- 0.45	41.56 +/- 0.64	44.09 +/- 0.45	52.16 +/- 1.49	45.44 +/- 1.45	27.23 +/- 1.75
CB / ROS + LA	44.09 +/- 0.45	41.56 +/- 0.64	44.09 +/- 0.45	52.16 +/- 1.49	45.44 +/- 1.45	27.23 +/- 1.75
CB / SMOTE	43.99 +/- 0.45	41.46 +/- 0.58	43.99 +/- 0.45	51.63 +/- 1.53	45.75 +/- 1.25	27.14 +/- 1.65
CB / SMOTE + LA	43.99 +/- 0.45	41.46 +/- 0.58	43.99 +/- 0.45	51.63 +/- 1.53	45.75 +/- 1.25	27.14 +/- 1.65
CB+EQ	44.78 +/- 0.44	42.35 +/- 0.56	44.78 +/- 0.44	52.34 +/- 1.42	46.57 +/- 1.31	28.30 +/- 1.59
CB+EQ + LA	45.73 +/- 0.44	43.83 +/- 0.57	45.73 +/- 0.44	54.63 +/- 1.60	46.98 +/- 1.39	30.06 +/- 1.79
CB+EQ / NCM	44.47 +/- 0.64	42.79 +/- 0.70	44.47 +/- 0.64	52.50 +/- 1.52	44.72 +/- 1.23	31.30 +/- 1.56
CB+EQ / natural	38.02 +/- 0.56	32.87 +/- 0.50	38.02 +/- 0.56	45.68 +/- 1.12	42.04 +/- 1.31	11.08 +/- 0.99
CB+EQ / natural + LA	44.20 +/- 0.59	45.33 +/- 0.56	44.20 +/- 0.59	57.63 +/- 2.23	48.02 +/- 1.25	30.50 +/- 1.87
CB+EQ / ROS	44.09 +/- 0.45	41.56 +/- 0.63	44.09 +/- 0.45	52.18 +/- 1.52	45.43 +/- 1.45	27.25 +/- 1.77
CB+EQ / ROS + LA	44.09 +/- 0.45	41.56 +/- 0.63	44.09 +/- 0.45	52.18 +/- 1.52	45.43 +/- 1.45	27.25 +/- 1.77
CB+EQ / SMOTE	44.00 +/- 0.46	41.48 +/- 0.56	44.00 +/- 0.46	51.63 +/- 1.54	45.79 +/- 1.29	27.16 +/- 1.56
CB+EQ / SMOTE + LA	44.00 +/- 0.46	41.48 +/- 0.56	44.00 +/- 0.46	51.63 +/- 1.54	45.79 +/- 1.29	27.16 +/- 1.56

CB+EQ identifies the REBAL adapter. /natural, /ROS and /SMOTE are retrained linear heads on that adapter. NCM uses fitted centroids. LA is a view of the same classifier, not another fit; balanced-head duplicate rows are deliberate. Zero-denominator class metrics use zero.

The accompanying CSVs retain every seed-level scalar (including worst-class F1 and head-tail gap), all per-class precision/recall/F1 values, the six-metric summaries, and all primary comparisons. No favorable subset replaces this complete table.

7. Complete results: DeiT-tiny | 10:1

Mean +/- sample SD across five paired seeds; values are percentages. Accuracy, macro-F1, macro recall and head/middle/tail macro-F1 are reported for every declared readout. Macro recall equals accuracy on this class-balanced evaluation set.

Method / readout	Accuracy	Macro F1	Macro recall	Head F1	Middle F1	Tail F1
CE	62.32 +/- 0.35	62.07 +/- 0.34	62.32 +/- 0.35	62.53 +/- 1.54	63.54 +/- 0.69	60.15 +/- 1.20
CE + LA	64.97 +/- 0.34	64.93 +/- 0.32	64.97 +/- 0.34	68.16 +/- 1.68	64.29 +/- 1.17	62.38 +/- 1.12
CE / NCM	62.91 +/- 0.26	62.82 +/- 0.30	62.91 +/- 0.26	65.29 +/- 1.46	62.19 +/- 0.84	61.01 +/- 1.74
CE / natural	62.07 +/- 0.45	61.89 +/- 0.45	62.07 +/- 0.45	62.47 +/- 1.58	63.17 +/- 0.81	60.04 +/- 1.16
CE / natural + LA	64.67 +/- 0.24	64.64 +/- 0.24	64.67 +/- 0.24	67.87 +/- 1.58	63.94 +/- 0.96	62.14 +/- 1.53
CE / ROS	64.53 +/- 0.27	64.43 +/- 0.27	64.53 +/- 0.27	66.71 +/- 1.47	63.69 +/- 0.87	62.91 +/- 1.40
CE / ROS + LA	64.53 +/- 0.27	64.43 +/- 0.27	64.53 +/- 0.27	66.71 +/- 1.47	63.69 +/- 0.87	62.91 +/- 1.40
CE / SMOTE	64.43 +/- 0.22	64.39 +/- 0.22	64.43 +/- 0.22	66.24 +/- 1.44	63.74 +/- 0.71	63.21 +/- 1.19
CE / SMOTE + LA	64.43 +/- 0.22	64.39 +/- 0.22	64.43 +/- 0.22	66.24 +/- 1.44	63.74 +/- 0.71	63.21 +/- 1.19
CB	64.76 +/- 0.33	64.60 +/- 0.35	64.76 +/- 0.33	66.21 +/- 1.42	64.11 +/- 0.88	63.51 +/- 1.41
CB + LA	65.00 +/- 0.32	64.88 +/- 0.32	65.00 +/- 0.32	67.16 +/- 1.36	64.11 +/- 1.12	63.42 +/- 1.38
CB / NCM	62.97 +/- 0.25	62.89 +/- 0.29	62.97 +/- 0.25	65.30 +/- 1.54	62.24 +/- 0.89	61.18 +/- 1.66
CB / natural	62.17 +/- 0.45	62.01 +/- 0.46	62.17 +/- 0.45	62.50 +/- 1.53	63.26 +/- 0.80	60.27 +/- 1.18
CB / natural + LA	64.70 +/- 0.22	64.67 +/- 0.22	64.70 +/- 0.22	67.95 +/- 1.61	63.97 +/- 1.01	62.15 +/- 1.45
CB / ROS	64.54 +/- 0.30	64.45 +/- 0.30	64.54 +/- 0.30	66.70 +/- 1.42	63.70 +/- 0.88	62.98 +/- 1.36
CB / ROS + LA	64.54 +/- 0.30	64.45 +/- 0.30	64.54 +/- 0.30	66.70 +/- 1.42	63.70 +/- 0.88	62.98 +/- 1.36
CB / SMOTE	64.44 +/- 0.17	64.40 +/- 0.15	64.44 +/- 0.17	66.26 +/- 1.38	63.68 +/- 0.78	63.30 +/- 1.16
CB / SMOTE + LA	64.44 +/- 0.17	64.40 +/- 0.15	64.44 +/- 0.17	66.26 +/- 1.38	63.68 +/- 0.78	63.30 +/- 1.16
CB+EQ	64.77 +/- 0.33	64.61 +/- 0.34	64.77 +/- 0.33	66.21 +/- 1.42	64.12 +/- 0.89	63.53 +/- 1.42
CB+EQ + LA	64.99 +/- 0.30	64.87 +/- 0.30	64.99 +/- 0.30	67.17 +/- 1.46	64.10 +/- 1.11	63.39 +/- 1.37
CB+EQ / NCM	62.95 +/- 0.25	62.87 +/- 0.29	62.95 +/- 0.25	65.26 +/- 1.52	62.22 +/- 0.91	61.17 +/- 1.66
CB+EQ / natural	62.16 +/- 0.47	61.99 +/- 0.48	62.16 +/- 0.47	62.51 +/- 1.56	63.26 +/- 0.79	60.22 +/- 1.21
CB+EQ / natural + LA	64.70 +/- 0.22	64.66 +/- 0.22	64.70 +/- 0.22	67.77 +/- 1.52	63.99 +/- 0.94	62.27 +/- 1.50
CB+EQ / ROS	64.52 +/- 0.29	64.42 +/- 0.30	64.52 +/- 0.29	66.67 +/- 1.40	63.68 +/- 0.87	62.94 +/- 1.32
CB+EQ / ROS + LA	64.52 +/- 0.29	64.42 +/- 0.30	64.52 +/- 0.29	66.67 +/- 1.40	63.68 +/- 0.87	62.94 +/- 1.32
CB+EQ / SMOTE	64.44 +/- 0.17	64.40 +/- 0.14	64.44 +/- 0.17	66.23 +/- 1.39	63.69 +/- 0.73	63.30 +/- 1.10
CB+EQ / SMOTE + LA	64.44 +/- 0.17	64.40 +/- 0.14	64.44 +/- 0.17	66.23 +/- 1.39	63.69 +/- 0.73	63.30 +/- 1.10

CB+EQ identifies the REBAL adapter. /natural, /ROS and /SMOTE are retrained linear heads on that adapter. NCM uses fitted centroids. LA is a view of the same classifier, not another fit; balanced-head duplicate rows are deliberate. Zero-denominator class metrics use zero.

The accompanying CSVs retain every seed-level scalar (including worst-class F1 and head-tail gap), all per-class precision/recall/F1 values, the six-metric summaries, and all primary comparisons. No favorable subset replaces this complete table.

8. Complete results: DeiT-tiny | 50:1

Mean +/- sample SD across five paired seeds; values are percentages. Accuracy, macro-F1, macro recall and head/middle/tail macro-F1 are reported for every declared readout. Macro recall equals accuracy on this class-balanced evaluation set.

Method / readout	Accuracy	Macro F1	Macro recall	Head F1	Middle F1	Tail F1
CE	51.98 +/- 0.30	49.44 +/- 0.69	51.98 +/- 0.30	55.58 +/- 1.02	57.43 +/- 0.70	35.40 +/- 2.71
CE + LA	59.89 +/- 0.68	60.11 +/- 0.48	59.89 +/- 0.68	67.18 +/- 2.01	61.01 +/- 0.85	52.24 +/- 2.28
CE / NCM	58.30 +/- 0.61	57.88 +/- 0.68	58.30 +/- 0.61	62.98 +/- 1.49	58.44 +/- 0.83	52.30 +/- 2.68
CE / natural	52.19 +/- 0.31	49.86 +/- 0.77	52.19 +/- 0.31	55.69 +/- 1.02	57.48 +/- 0.90	36.50 +/- 2.92
CE / natural + LA	59.59 +/- 0.58	59.92 +/- 0.41	59.59 +/- 0.58	67.06 +/- 2.16	60.98 +/- 0.91	51.83 +/- 2.51
CE / ROS	59.43 +/- 0.48	58.96 +/- 0.61	59.43 +/- 0.48	63.73 +/- 1.83	59.94 +/- 0.83	53.29 +/- 2.50
CE / ROS + LA	59.43 +/- 0.48	58.96 +/- 0.61	59.43 +/- 0.48	63.73 +/- 1.83	59.94 +/- 0.83	53.29 +/- 2.50
CE / SMOTE	59.42 +/- 0.42	59.05 +/- 0.56	59.42 +/- 0.42	63.12 +/- 1.74	60.13 +/- 0.91	53.96 +/- 2.54
CE / SMOTE + LA	59.42 +/- 0.42	59.05 +/- 0.56	59.42 +/- 0.42	63.12 +/- 1.74	60.13 +/- 0.91	53.96 +/- 2.54
CB	59.64 +/- 0.52	58.98 +/- 0.60	59.64 +/- 0.52	63.20 +/- 1.70	60.37 +/- 0.71	53.43 +/- 2.26
CB + LA	60.29 +/- 0.49	59.85 +/- 0.57	60.29 +/- 0.49	65.10 +/- 1.76	60.38 +/- 0.86	54.14 +/- 2.22
CB / NCM	58.55 +/- 0.63	58.17 +/- 0.69	58.55 +/- 0.63	63.13 +/- 1.61	58.53 +/- 0.91	52.94 +/- 2.73
CB / natural	52.50 +/- 0.34	50.33 +/- 0.77	52.50 +/- 0.34	55.82 +/- 0.99	57.59 +/- 0.74	37.67 +/- 2.95
CB / natural + LA	59.75 +/- 0.60	60.10 +/- 0.48	59.75 +/- 0.60	67.00 +/- 2.04	61.07 +/- 1.05	52.34 +/- 2.51
CB / ROS	59.50 +/- 0.48	59.04 +/- 0.62	59.50 +/- 0.48	63.71 +/- 1.72	59.81 +/- 0.82	53.67 +/- 2.50
CB / ROS + LA	59.50 +/- 0.48	59.04 +/- 0.62	59.50 +/- 0.48	63.71 +/- 1.72	59.81 +/- 0.82	53.67 +/- 2.50
CB / SMOTE	59.47 +/- 0.49	59.10 +/- 0.61	59.47 +/- 0.49	63.05 +/- 1.78	60.11 +/- 0.77	54.21 +/- 2.40
CB / SMOTE + LA	59.47 +/- 0.49	59.10 +/- 0.61	59.47 +/- 0.49	63.05 +/- 1.78	60.11 +/- 0.77	54.21 +/- 2.40
CB+EQ	59.62 +/- 0.53	58.96 +/- 0.61	59.62 +/- 0.53	63.18 +/- 1.72	60.36 +/- 0.70	53.39 +/- 2.29
CB+EQ + LA	60.28 +/- 0.49	59.84 +/- 0.57	60.28 +/- 0.49	65.07 +/- 1.76	60.38 +/- 0.84	54.16 +/- 2.22
CB+EQ / NCM	58.51 +/- 0.60	58.14 +/- 0.67	58.51 +/- 0.60	63.13 +/- 1.61	58.48 +/- 0.90	52.90 +/- 2.69
CB+EQ / natural	52.49 +/- 0.34	50.32 +/- 0.76	52.49 +/- 0.34	55.82 +/- 0.98	57.59 +/- 0.73	37.63 +/- 2.92
CB+EQ / natural + LA	59.76 +/- 0.60	60.11 +/- 0.48	59.76 +/- 0.60	67.00 +/- 2.04	61.10 +/- 1.05	52.32 +/- 2.51
CB+EQ / ROS	59.48 +/- 0.49	59.03 +/- 0.63	59.48 +/- 0.49	63.72 +/- 1.72	59.79 +/- 0.81	53.65 +/- 2.49
CB+EQ / ROS + LA	59.48 +/- 0.49	59.03 +/- 0.63	59.48 +/- 0.49	63.72 +/- 1.72	59.79 +/- 0.81	53.65 +/- 2.49
CB+EQ / SMOTE	59.49 +/- 0.51	59.12 +/- 0.63	59.49 +/- 0.51	63.08 +/- 1.80	60.11 +/- 0.80	54.23 +/- 2.42
CB+EQ / SMOTE + LA	59.49 +/- 0.51	59.12 +/- 0.63	59.49 +/- 0.51	63.08 +/- 1.80	60.11 +/- 0.80	54.23 +/- 2.42

CB+EQ identifies the REBAL adapter. /natural, /ROS and /SMOTE are retrained linear heads on that adapter. NCM uses fitted centroids. LA is a view of the same classifier, not another fit; balanced-head duplicate rows are deliberate. Zero-denominator class metrics use zero.

The accompanying CSVs retain every seed-level scalar (including worst-class F1 and head-tail gap), all per-class precision/recall/F1 values, the six-metric summaries, and all primary comparisons. No favorable subset replaces this complete table.

9. Complete results: DeiT-tiny | 100:1

Mean +/- sample SD across five paired seeds; values are percentages. Accuracy, macro-F1, macro recall and head/middle/tail macro-F1 are reported for every declared readout. Macro recall equals accuracy on this class-balanced evaluation set.

Method / readout	Accuracy	Macro F1	Macro recall	Head F1	Middle F1	Tail F1
CE	46.17 +/- 0.34	41.47 +/- 0.51	46.17 +/- 0.34	52.37 +/- 1.25	53.46 +/- 0.56	18.75 +/- 2.33
CE + LA	55.63 +/- 0.37	56.37 +/- 0.37	55.63 +/- 0.37	66.56 +/- 1.91	59.76 +/- 1.22	42.93 +/- 2.13
CE / NCM	55.01 +/- 0.55	54.10 +/- 0.67	55.01 +/- 0.55	61.43 +/- 1.63	55.96 +/- 0.65	45.02 +/- 2.93
CE / natural	46.78 +/- 0.33	42.47 +/- 0.41	46.78 +/- 0.33	52.81 +/- 1.31	53.84 +/- 0.61	20.91 +/- 1.97
CE / natural + LA	55.49 +/- 0.46	56.23 +/- 0.52	55.49 +/- 0.46	66.26 +/- 1.70	59.42 +/- 1.36	43.17 +/- 2.16
CE / ROS	55.84 +/- 0.69	54.70 +/- 0.91	55.84 +/- 0.69	61.83 +/- 1.61	57.06 +/- 1.14	45.31 +/- 2.70
CE / ROS + LA	55.84 +/- 0.69	54.70 +/- 0.91	55.84 +/- 0.69	61.83 +/- 1.61	57.06 +/- 1.14	45.31 +/- 2.70
CE / SMOTE	55.98 +/- 0.68	54.91 +/- 0.91	55.98 +/- 0.68	61.24 +/- 1.60	57.68 +/- 1.05	45.92 +/- 2.88
CE / SMOTE + LA	55.98 +/- 0.68	54.91 +/- 0.91	55.98 +/- 0.68	61.24 +/- 1.60	57.68 +/- 1.05	45.92 +/- 2.88
CB	55.55 +/- 0.41	53.98 +/- 0.63	55.55 +/- 0.41	60.90 +/- 1.44	57.78 +/- 0.97	43.37 +/- 2.56
CB + LA	56.58 +/- 0.46	55.39 +/- 0.61	56.58 +/- 0.46	63.34 +/- 1.45	57.92 +/- 0.96	45.01 +/- 2.42
CB / NCM	55.36 +/- 0.54	54.55 +/- 0.68	55.36 +/- 0.54	61.56 +/- 1.69	56.09 +/- 0.79	46.09 +/- 3.01
CB / natural	47.27 +/- 0.35	43.21 +/- 0.41	47.27 +/- 0.35	53.00 +/- 1.36	54.19 +/- 0.61	22.58 +/- 2.11
CB / natural + LA	56.00 +/- 0.45	56.79 +/- 0.46	56.00 +/- 0.45	66.25 +/- 1.83	59.62 +/- 1.26	44.66 +/- 2.20
CB / ROS	55.96 +/- 0.63	54.80 +/- 0.86	55.96 +/- 0.63	61.83 +/- 1.68	57.07 +/- 1.14	45.62 +/- 2.70
CB / ROS + LA	55.96 +/- 0.63	54.80 +/- 0.86	55.96 +/- 0.63	61.83 +/- 1.68	57.07 +/- 1.14	45.62 +/- 2.70
CB / SMOTE	56.03 +/- 0.64	54.93 +/- 0.90	56.03 +/- 0.64	61.18 +/- 1.70	57.65 +/- 1.03	46.04 +/- 3.00
CB / SMOTE + LA	56.03 +/- 0.64	54.93 +/- 0.90	56.03 +/- 0.64	61.18 +/- 1.70	57.65 +/- 1.03	46.04 +/- 3.00
CB+EQ	55.53 +/- 0.42	53.95 +/- 0.65	55.53 +/- 0.42	60.86 +/- 1.41	57.76 +/- 0.97	43.34 +/- 2.60
CB+EQ + LA	56.55 +/- 0.47	55.35 +/- 0.62	56.55 +/- 0.47	63.30 +/- 1.43	57.89 +/- 0.98	44.98 +/- 2.45
CB+EQ / NCM	55.34 +/- 0.53	54.52 +/- 0.67	55.34 +/- 0.53	61.54 +/- 1.71	56.06 +/- 0.79	46.07 +/- 2.98
CB+EQ / natural	47.26 +/- 0.34	43.19 +/- 0.42	47.26 +/- 0.34	53.01 +/- 1.34	54.18 +/- 0.65	22.52 +/- 2.12
CB+EQ / natural + LA	55.99 +/- 0.48	56.79 +/- 0.47	55.99 +/- 0.48	66.24 +/- 1.80	59.61 +/- 1.24	44.65 +/- 2.21
CB+EQ / ROS	55.97 +/- 0.62	54.81 +/- 0.85	55.97 +/- 0.62	61.83 +/- 1.69	57.09 +/- 1.12	45.62 +/- 2.72
CB+EQ / ROS + LA	55.97 +/- 0.62	54.81 +/- 0.85	55.97 +/- 0.62	61.83 +/- 1.69	57.09 +/- 1.12	45.62 +/- 2.72
CB+EQ / SMOTE	56.00 +/- 0.65	54.90 +/- 0.91	56.00 +/- 0.65	61.17 +/- 1.70	57.63 +/- 1.08	45.98 +/- 2.99
CB+EQ / SMOTE + LA	56.00 +/- 0.65	54.90 +/- 0.91	56.00 +/- 0.65	61.17 +/- 1.70	57.63 +/- 1.08	45.98 +/- 2.99

CB+EQ identifies the REBAL adapter. /natural, /ROS and /SMOTE are retrained linear heads on that adapter. NCM uses fitted centroids. LA is a view of the same classifier, not another fit; balanced-head duplicate rows are deliberate. Zero-denominator class metrics use zero.

The accompanying CSVs retain every seed-level scalar (including worst-class F1 and head-tail gap), all per-class precision/recall/F1 values, the six-metric summaries, and all primary comparisons. No favorable subset replaces this complete table.

10. Paired encoder effects

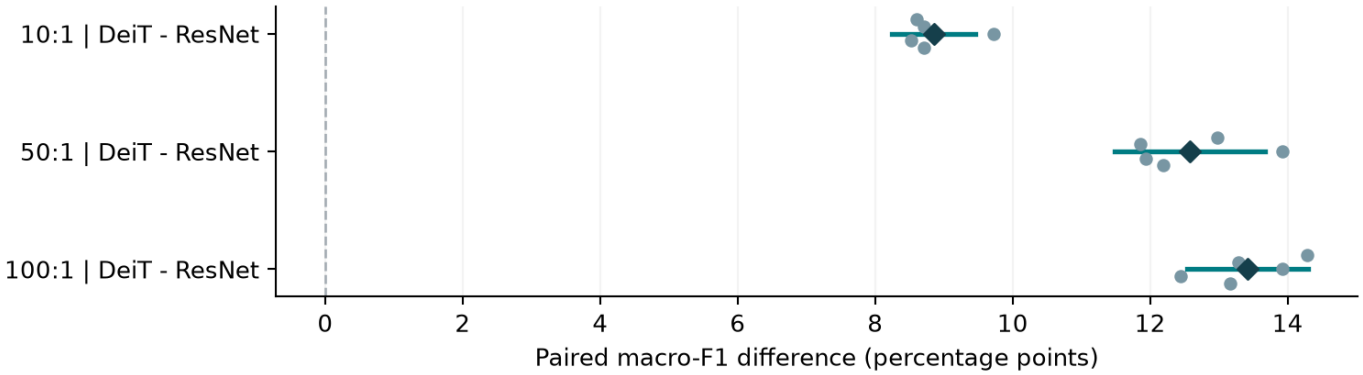


Figure 1. DeiT-minus-ResNet effects under the prespecified CB+EQ plus SMOTE configuration. Points show five paired seed differences; diamonds show means and bars unadjusted paired t 95% intervals. Positive values favor DeiT.

Paired contrast	Delta (pp)	95% t interval	t p, Holm	Exact p, Holm
10:1 | DeiT - ResNet	+8.85	[+8.24, +9.46]	3.13e-05	0.938
50:1 | DeiT - ResNet	+12.58	[+11.49, +13.67]	7.30e-05	0.938
100:1 | DeiT - ResNet	+13.42	[+12.53, +14.30]	2.85e-05	0.938

The pairing shares calibration/fit row identities, rank assignment and classifier initialization across encoders. Each seed changes both optimization and the shared split/rank construction. These are five paired replications on one dataset, not 10,000 independent training replications.

Holm correction covers all 15 primary comparisons, separately for paired t and exact sign-flip p-values. Confidence intervals are not multiplicity-adjusted. With five pairs, the exact two-sided p-value floor is 2/32=0.0625 before correction. A t-test can yield a smaller p-value only under its distributional assumptions, which are difficult to assess with five seeds. Nonsignificance does not establish equivalence.

The encoder effect combines learned pretraining weights, original recipe, architecture, capacity and adapter input width. Matching local data and downstream schedules does not isolate any one of those causes.

11. Paired mechanism effects

Paired contrast	Delta (pp)	95% t interval	t p, Holm	Exact p, Holm
DeiT-tiny 10:1 | CB+EQ - CB	-0.01	[-0.04, +0.01]	1.000	1.000
DeiT-tiny 10:1 | SMOTE - ROS	-0.02	[-0.25, +0.22]	1.000	1.000
ResNet18 10:1 | CB+EQ - CB	-0.02	[-0.04, +0.00]	0.825	1.000
ResNet18 10:1 | SMOTE - ROS	-0.12	[-0.47, +0.23]	1.000	1.000
DeiT-tiny 50:1 | CB+EQ - CB	-0.01	[-0.04, +0.01]	1.000	1.000
DeiT-tiny 50:1 | SMOTE - ROS	+0.09	[-0.20, +0.39]	1.000	1.000
ResNet18 50:1 | CB+EQ - CB	-0.00	[-0.04, +0.03]	1.000	1.000
ResNet18 50:1 | SMOTE - ROS	-0.08	[-0.22, +0.05]	1.000	1.000
DeiT-tiny 100:1 | CB+EQ - CB	-0.02	[-0.04, +0.00]	0.919	1.000
DeiT-tiny 100:1 | SMOTE - ROS	+0.08	[-0.07, +0.24]	1.000	1.000
ResNet18 100:1 | CB+EQ - CB	-0.01	[-0.04, +0.02]	1.000	1.000
ResNet18 100:1 | SMOTE - ROS	-0.09	[-0.27, +0.09]	1.000	1.000

SMOTE-minus-ROS holds the CB+EQ representation fixed and compares fresh within-class interpolation with class-uniform real sampling. CB+EQ-minus-CB uses natural classifier retraining to isolate equalization within the declared class-balanced objective and schedule. These 12 contrasts share the same 15-member family with the three encoder contrasts.

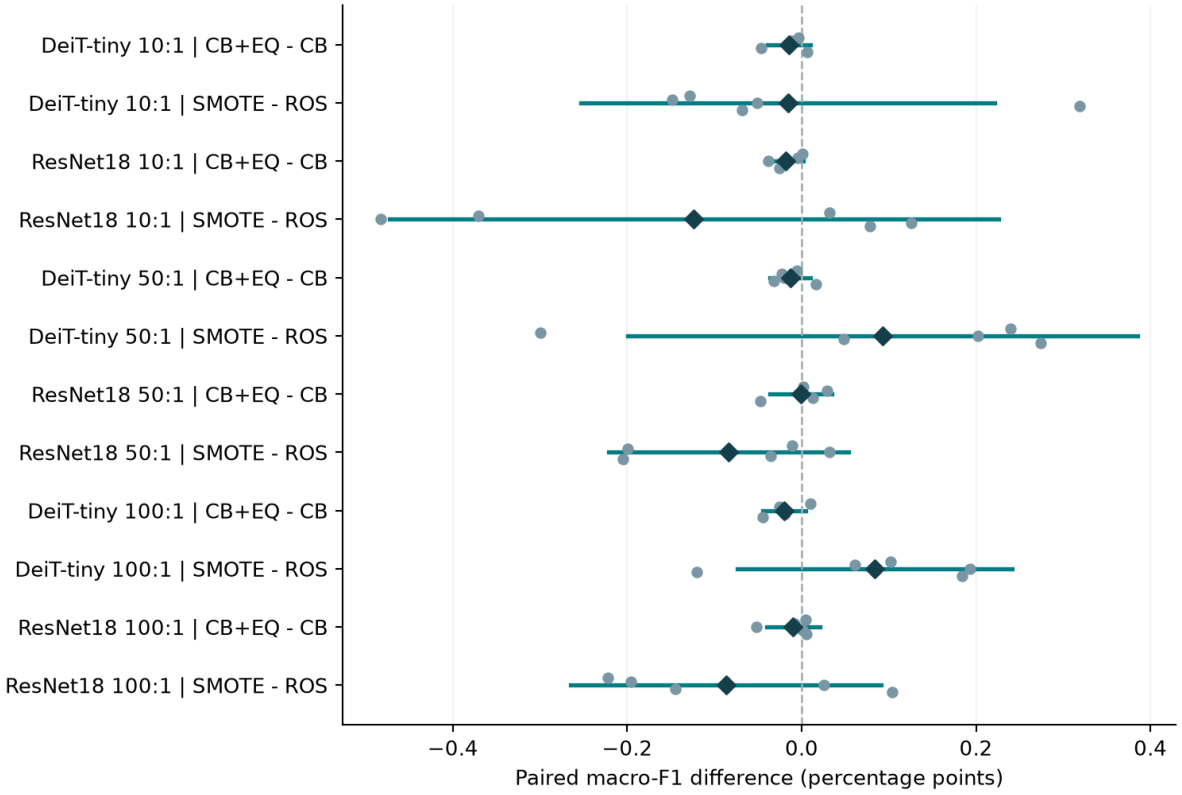


Figure 2. All 12 paired mechanism effects, retaining negative and inconsistent outcomes. Points are seeds; diamonds and bars show mean and unadjusted t interval. A positive interpolation difference is not evidence that the resulting pipeline outperforms all native, centroid or adjusted controls.

12. Absolute patterns and exploratory geometry

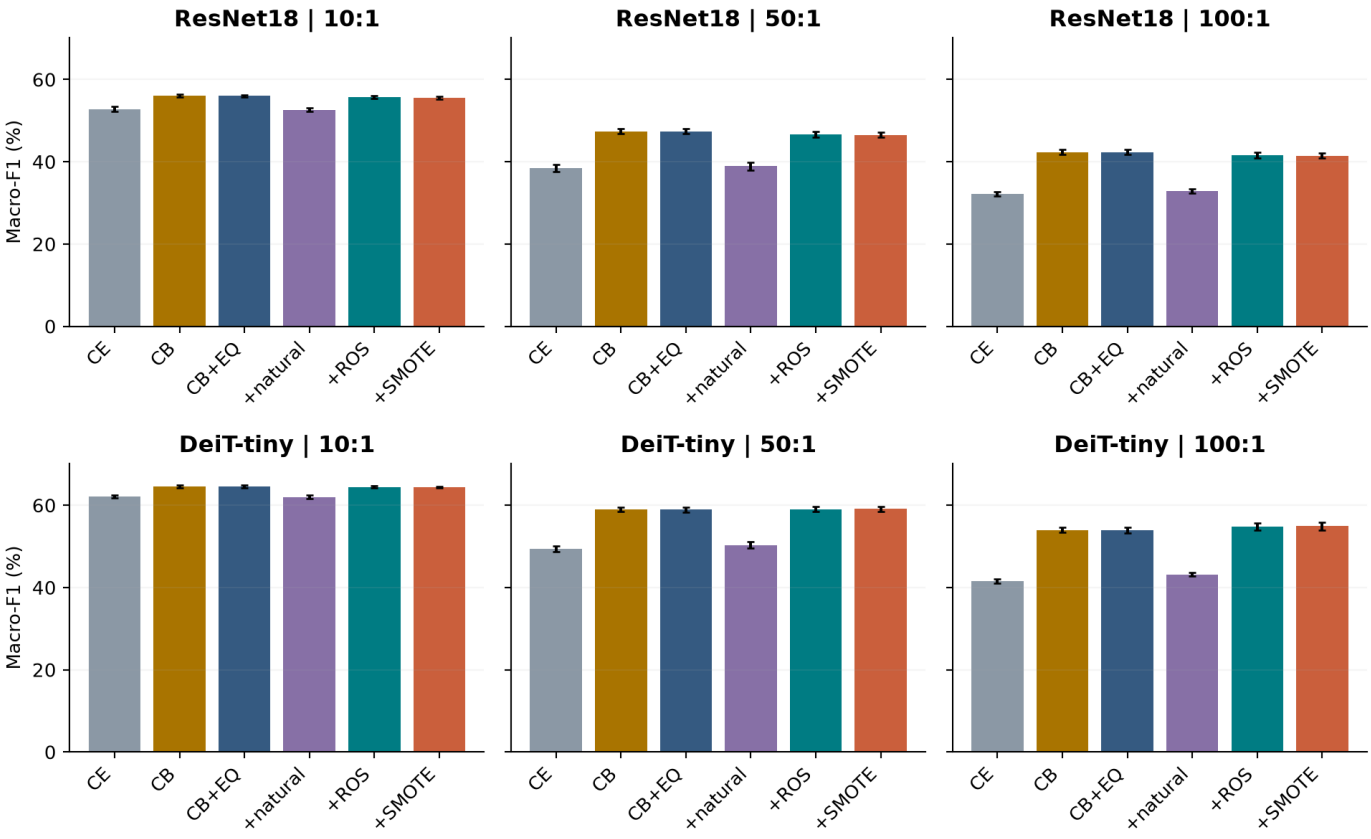
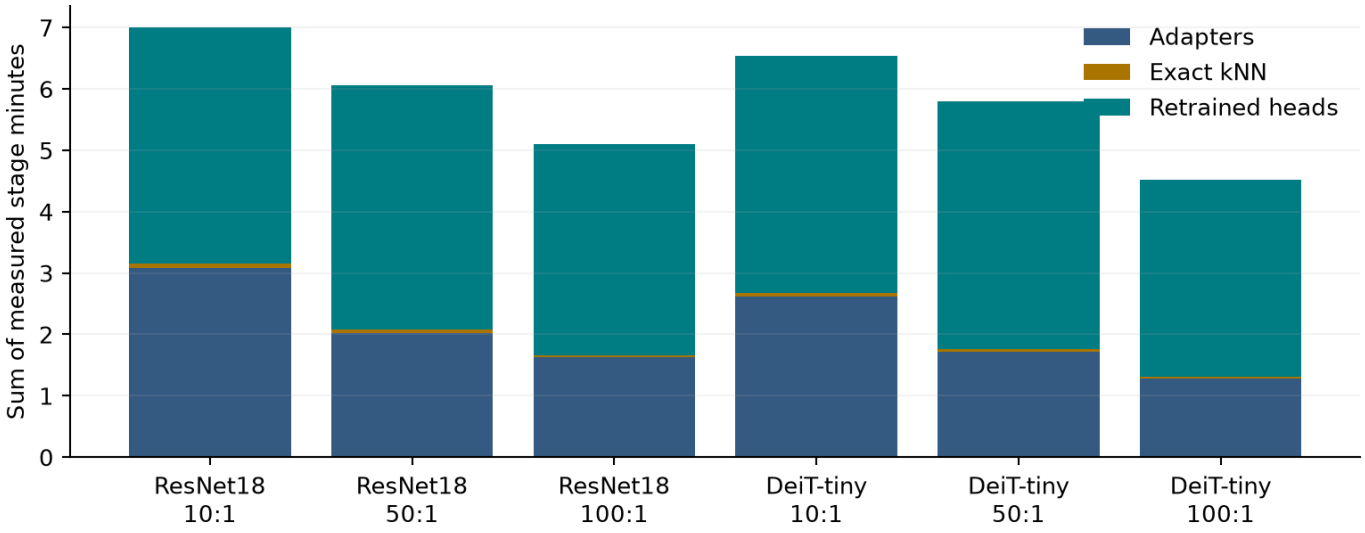


Figure 3. Native CE, CB and CB+EQ controls and natural/ROS/SMOTE retraining on CB+EQ features. Bars show mean macro-F1 and error bars show SD. All remaining LA/NCM views are in the full tables. The y-axis starts at zero.

Encoder / ratio	CB+EQ NCM - CB NCM	Active EQ eligibility (mean / minimum)
ResNet18 | 10:1	-0.06 pp	82.5% / 35.6%
ResNet18 | 50:1	-0.05 pp	64.8% / 4.1%
ResNet18 | 100:1	-0.05 pp	58.6% / 0.0%
DeiT-tiny | 10:1	-0.02 pp	82.5% / 35.6%
DeiT-tiny | 50:1	-0.03 pp	64.8% / 4.1%
DeiT-tiny | 100:1	-0.02 pp	58.6% / 0.0%

NCM differences are exploratory representation evidence, outside the primary inference family. EQ eligibility is the active-epoch batch fraction containing at least two instances of a class. Its minimum spans seed/class pairs. Rare-class exposure may limit regularization; exposure alone does not prove improvement.

13. Actual runtime and resource accounting



Encoder / ratio	Adapter min	Exact kNN min	Head min
ResNet18 | 10:1	3.09	0.06	3.85
ResNet18 | 50:1	2.03	0.04	3.99
ResNet18 | 100:1	1.63	0.03	3.44
DeiT-tiny | 10:1	2.62	0.05	3.87
DeiT-tiny | 50:1	1.71	0.05	4.04
DeiT-tiny | 100:1	1.28	0.03	3.21

Figure 4. Sum of measured stage durations over five seeds and three objectives. Concurrent worker sums are not elapsed wall time. Separate encoding of calibration/test features, prediction compression, input checking, scheduling and verification are not all included in optimizer/graph timers. Resume timers can omit discarded work before the last checkpoint.

Frozen feature extraction	Inference minutes	Feature dimension
ResNet18	2.76	512
DeiT-tiny	2.67	192

Study-plan creation to the last completed cell spans 13.42 minutes. This includes scheduling and may contain idle time; it is not an isolated kernel benchmark. Feature extraction above records inference time separately for 110,000 images per encoder.

Execution uses CPU, four PyTorch threads per worker, and exact CPU within-class neighbors. Train-only CPU/MPS pilots selected the backend before comparison scores. Precomputed encoder features were extracted on MPS. Exact neighbor computation remains quadratic within each class, with bounded memory.

Recorded downstream software: imbalanced-learn 0.14.2, numpy 2.5.4, scikit-learn 1.9.1, scipy 1.18.1, torch 2.14.1. Deterministic-algorithm warnings and saved RNG state improve reproducibility but do not guarantee cross-platform bitwise identity.

14. Independent evidence and reproducibility

The independent verifier recomputed test metrics from saved logits and labels, reconstructed all calibration LA curves, rebuilt every declared source-row split and fit-only scaler, checked training budgets and graph integrity, and recomputed all summary statistics and 15 paired comparisons. It independently parsed source ZIP labels/order and checked archive, cache, checkpoint and other artifact hashes.

Verified item	Scope
Complete grid	30/30 cells; 810 readouts; no missing cases
Calibration	360 fixed-grid curves; balanced aliases reconstructed
Aggregate	162 complete five-seed method summaries; one 15-contrast family
Pairing	Identical raw split hash and initial head hash per encoder pair
Source integrity	Pinned official weights, cache arrays, source modules and original archive
Limits	Frozen encoder inference and pickle checkpoint internals are not re-executed; stored logits are checked for consistency.

The report builder refuses partial, pilot, stale, changed-source or changed-budget evidence. It rechecks current source, cache, cell artifact and aggregate hashes against the independently verified snapshot before authoring any result artifact. Numerical consistency is not cryptographic authentication of an untrusted run and does not imply benchmark superiority.

Before launch the full repository suite passed 81 tests, including nine verifier tests. Targeted runner tests cover sampler/RNG equivalence, exact interrupted resume, matching initialization, artifact tamper rejection and exclusion of heldout model calls from throughput pilots. Reproduce the portable checks with `python -m unittest discover -s tests -v`.

Published code commit: `5a25efab7909d3dab51a586afef8f5f477ba0306`.

Protocol fingerprint

```
d3df8ea7cd7484045aadc71d7e243d09b3cc9270fd9efd27d6f4b32c4c620558
```

Runner SHA256

```
0520feeca4505c6a0dc0e012b9d02867237b96cb65fbf44034f9af95ef23fdd8
```

Independent verifier SHA256

```
fabd98308cb5df11640cefbfa5484ebf3586485d4a7ff15ba9c241b1e85f1c7a
```

Verified aggregate SHA256

```
86dcd6c4cdf916975b6ea438168f950e5b3f459516e86d48cc40f0dc6879fd4b
```

Portable reproduction is documented in docs/vit_rebal_experiment.md. CSV values use fractions; report tables/plots use percentages or percentage-point deltas. The complete aggregate JSON accompanies the report. Private operational records, credentials, raw source images and model download caches are excluded from report publication.

15. Limits and primary references

Pretraining overlap

Both encoders were pretrained on ImageNet, while Tiny ImageNet derives from ImageNet. Local deduplication cannot remove unknown pretraining image/label exposure. This transfer comparison cannot establish clean unseen-image generalization.

Encoder-package confounding

Feature width, frozen parameter count, original recipe and architecture differ. Shared bicubic preprocessing controls input convention but is not ResNet18 V1's default. No architecture-only causal effect or tuned state-of-the-art claim follows.

Adapted REBAL scope

The trained object is a small feature adapter with a downstream classifier. Frozen DeiT features, streaming latent SMOTE, count-aware CB and deferred weighting do not reproduce end-to-end ViT training or a GAN. Historical CIFAR GAN and earlier native64/ResNet32 studies are separate experiments.

Statistical resolution

Five seeds change local split/rank and optimization together. Ratios share source material and are correlated. Exact-test resolution is coarse; t intervals assume a seed-difference distribution that is hard to validate. Confidence intervals do not cover new datasets, recipes or architectures.

Controls and negative findings

A positive SMOTE-minus-ROS contrast does not establish dominance over CE, native CB, NCM or calibrated heads. Full tables preserve all controls and negative effects. LA aliases are named readouts, not extra training replications; NCM/EQ exposure analyses remain exploratory.

Primary sources

1. Touvron et al. (2021). Training data-efficient image transformers & distillation through attention. ICML.

2. Official DeiT repository: non-distilled pretrained models and evaluation transforms.

3. Torchvision ResNet18 weights and default preprocessing documentation.

4. Cui et al. (2019). Class-Balanced Loss Based on Effective Number of Samples.

5. Kang et al. (2020). Decoupling Representation and Classifier for Long-Tailed Recognition.

6. Chawla et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique.

7. Menon et al. (2021). Long-tail Learning via Logit Adjustment.